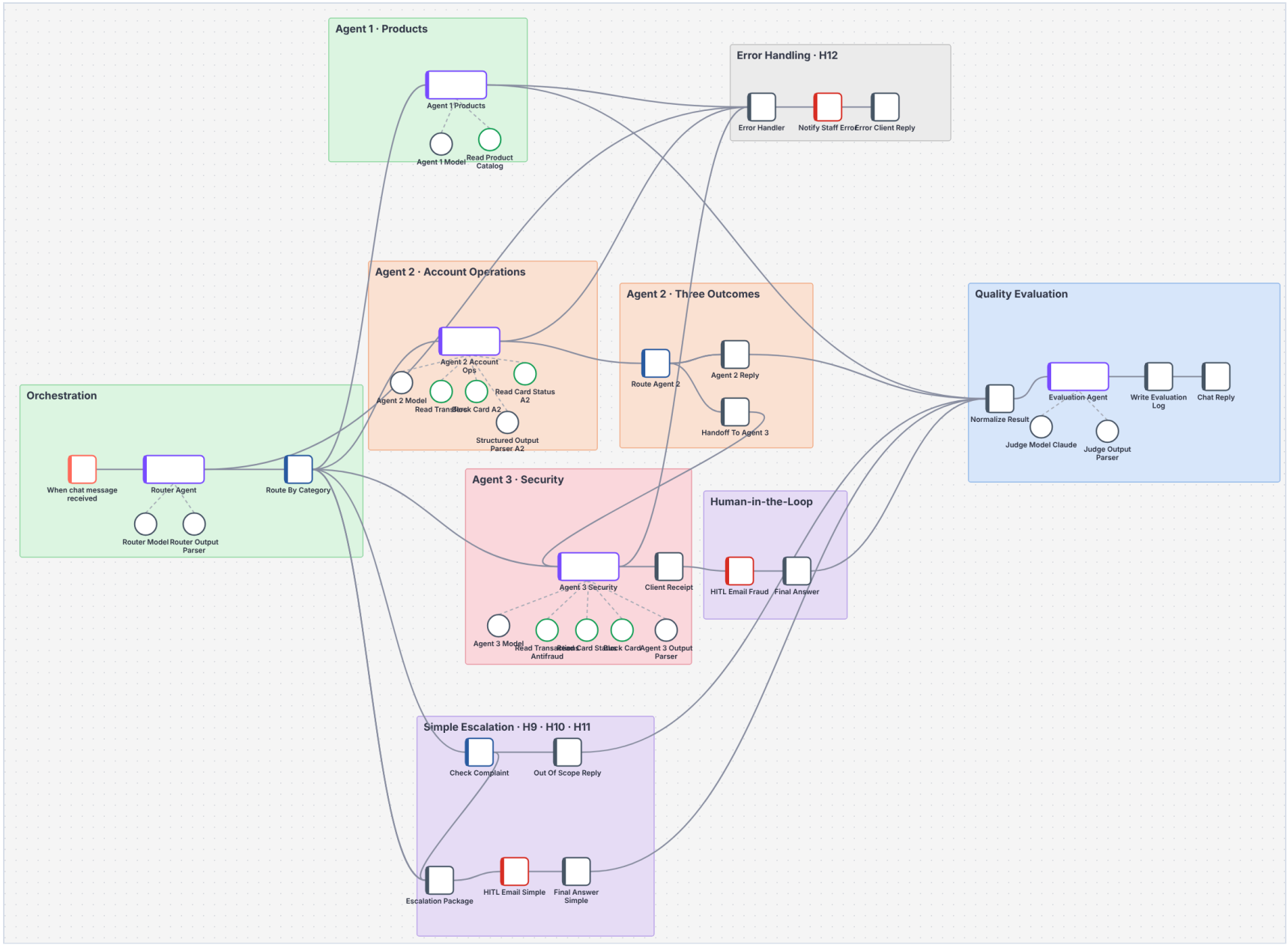
Multi-agent customer service for a bank, built in n8n

Alexej Kotelnikov | Project from the AI Business Automation Specialist training, Beam Institute of Technology, August 2026 | test data



Workflow canvas: AI agents routing and checks data sources (Google Sheets) human-in-the-loop (Gmail) trigger dashed lines: models, tools and output parsers attached to each agent

What the system does and how it is built

5 AI agents, 1 human approval point, 5 data sources, automatic quality evaluation of every answer

ARCHITECTURE

Router agent	Classifies each inquiry into 5 categories with confidence and complaint flag. A code switch checks the result against a fixed list, anything unexpected goes to a human.
Agent 1 Products	Answers product and fee questions from the catalogue. Fully autonomous, no access to accounts.
Agent 2 Account ops	Investigates transfers. Sees the antifraud flag but not the score behind it. On a hold it blocks the card and hands the case to Agent 3.
Agent 3 Security	Handles security incidents and prepares a package for staff. Acts on the merits only after human approval.
Human-in-the-loop	Staff confirm, edit or replace the agent's draft by email.
Evaluation agent	Claude as independent judge (agents run on GPT-4o): 7 criteria, scale 1 to 5, threshold 3.5, cases below go to review.
Error handling	All agent errors converge in one handler: apology to the customer, email to staff.

TOOLS

n8n, AI agents with structured output parsers, OpenRouter (GPT-4o), Claude, Google Sheets as data sources, Gmail, n8n data table as evaluation log.

RESULTS FROM TESTING

19
test inquiries, all 5 categories and all 3 staff actions covered

4.4 / 5
average quality score from the judge

3.9 → 4.9
same case: agent draft vs. after human edit

- The judge found a real design defect (an internal staff package leaking to the customer) before a human noticed it.

DESIGN PRINCIPLES

- **The more irreversible and visible an action, the lower the agent's autonomy.** Product info is autonomous, card blocking is autonomous with notification, refunds need a human.
- **The model decides, code checks.** Routing and audit fields are validated by code, not filled by the model.
- **Rights are separated by data, not by instructions.** Agents get different tools and tables, so a prompt cannot talk them out of their limits.
- **Agents coordinate through shared state.** Agent 3 re-reads the card status before acting, so a card is never blocked twice.

LIMITS AND NEXT STEPS

A training system, not production. 7 of 12 human handover conditions are implemented. Next: webhook delivery of the final answer, counters for follow-up questions, and a weekly quality report for the team lead.

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